



Analysis of Counter-Attacks in Elite Football in collaboration with DBU

From Ball Recovery to Offensive Advantage

Bachelor Thesis

University of Copenhagen

Department of Computer Science

Mikkel Keller Andreassen

Supervisor: Jon Sparring

External collaborator: Lasse Meinert Pedersen (DBU)

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Preface

This bachelor thesis was prepared at the Department of Computer Science at the University of Copenhagen in collaboration with the Danish Football Association (DBU).

The project investigates how synchronized event and tracking data can be used to define, detect, and analyse counter-attacks in elite football through a combination of rule-based methods, human annotation, and machine learning.

I would like to thank my supervisor, Jon Sparring, for guidance and feedback throughout the project. I would also like to thank Lasse Meinert Pedersen and DBU for providing access to the data and for valuable discussions related to football analysis and practical applications of the work.

Finally, I would like to thank the seven volunteer raters who contributed to the manual annotation process used in the human-guided evaluation of the models.

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Mikkel Keller Andreassen

Trq281@alumni.ku.dk

Abstract

Counter-attacks are an important part of modern football, but they are difficult to define consistently from event and tracking data because they depend on both tactical context and temporal development. This project develops a reproducible framework for detecting and analysing counter-attacks using synchronized event and tracking data from DBU national-team matches.

First, a rule-based definition was constructed by combining event-based possession regains with tracking-based spatial measurements and distance-scaled thresholds for duration, pass count, speed, in-play time, and territorial progression. The definition was validated against human annotations and showed reasonable agreement with human football judgment. Second, an XGBoost classifier was trained on engineered event and tracking features to model counter-attack detection from a broader candidate pool of regain situations. A hybrid human-guided version was then developed by incorporating majority-vote human labels and agreement-based sample weights. This hybrid model achieved the strongest agreement with the available human-labeled evaluation set, outperforming both the rule-based definition and a rule-trained XGBoost baseline.

Finally, the framework was applied to analyse counter-attacking patterns across Danish national teams. The results suggest that the Danish men’s senior team did not primarily lack counter-attacking volume, but rather final outcome quality, as their attacks less often reached dangerous end locations compared with tournament benchmarks. The project demonstrates that counter-attacks can be extracted, validated, and analysed reproducibly through a combination of rule-based logic, human annotation, and machine learning. However, the findings should be interpreted in light of limitations related to the small human-labeled evaluation set, event annotation dependency, and the difficulty of capturing tactical intention from observable data alone.

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1 Introduction

1.1 Motivation

A ball recovery in defensive midfield. A quick forward pass. Three seconds later, the ball is in the opponent’s half before the defence has fully reorganised. These brief transition moments are among the most dangerous phases in modern football and are often associated with high-quality scoring opportunities. In football analysis, such situations are commonly referred to as counter-attacks.

Counter-attacks exploit temporary defensive imbalances immediately following a regain of possession and are therefore considered an important component of modern elite football. At the same time, the increasing availability of high-frequency tracking and event data enables a more structured and quantitative analysis of these transition phases.

1.2 Background on Counter-attacks

In football, match play alternates between organized possession phases and transitional moments following a change of possession. Counter-attacks occur immediately after regaining possession and are characterized by rapid forward progression, dynamic spatial reorganization, and temporary numerical or positional advantages.

From an analytical perspective, however, counter-attacks are difficult to define formally. Tactical interpretations vary, and concepts such as “speed”, “directness”, and “defensive imbalance” must be translated into measurable quantities derived from event and tracking data. This creates a methodological challenge: how to construct a reproducible definition that captures the essence of transitional attacking play while remaining grounded in observable data.

1.3 Why Counter-attacks Matter

Evidence from elite football consistently shows that counter-attacks are relatively infrequent but highly effective attacking situations. In a sample of 1,971 possessions from English Premier League matches, only 9.5% of possessions were classified as counter-attacks, yet 24.9% of these produced a scoring opportunity, compared with 7.5% for combinative attacks and 1.6% for direct attacks (González-Rodenas et al. 2020). Similar findings have been reported across multiple competitions and contexts, where counter-attacks and fast attacks consistently outperform positional attacks in terms of offensive efficiency and chance creation (Tenga et al. 2010; Sarmiento et al. 2018). For example, Tenga et al. (2010) found that 52% of goal-scoring possessions in Norwegian professional football originated from counter-attacks, while Sarmiento et al. (2018) reported that both counter-attacks and

fast attacks were significantly more effective than positional attacks across top European leagues and the UEFA Champions League.

Recent studies further indicate that successful counter-attacks are typically characterized by rapid progression and short sequence duration. Analyses from La Liga and international tournament football show that counter-attacks most often develop within approximately 6–15 seconds and are strongly associated with dangerous outcomes such as shots and goals (Michailidis et al. 2024; FIFA Training Centre 2023). Large-scale transition research additionally underlines the broader strategic importance of offensive transitions in modern elite football (Eusebio et al. 2025; Yang et al. 2025).

1.4 Research Gap and Aim

Despite the tactical importance of counter-attacks, practical analysis still faces a core methodological challenge: identifying counter-attacking sequences reliably and reproducibly from large event and tracking datasets. Existing approaches often rely either on manual annotation or on rule-based pipelines whose validity depends strongly on data structure, threshold selection, and contextual interpretation (Yang et al. 2025). As a result, there remains a need for analytical frameworks that combine reproducibility, interpretability, and alignment with human football understanding.

2 Problem Statement and Scope

2.1 Problem formulation

The primary objective of this project is to develop a reproducible framework for detecting and analysing counter-attacks from synchronized event and tracking data in elite football. Counter-attacks are studied as structured transition phases following a regain of possession, and offensive success is evaluated through measurable outcomes such as shots, box entries, and final-third entries.

To address this objective, the project first develops a transparent rule-based definition of counter-attacks using event-based regain information combined with tracking-based spatial and temporal measurements. The purpose of this framework is to establish a reproducible and tactically meaningful method for identifying counter-attacking situations in large-scale football datasets.

The project then investigates whether this counter-attack concept can be learned and refined through supervised machine learning. In particular, a hybrid human-guided XGBoost model is developed by combining rule-generated labels with manually annotated human labels. The aim is not to replace the rule-based framework entirely, but to evaluate

whether a data-driven model can improve alignment with human football judgment while preserving reproducibility and scalability.

Finally, the detected counter-attacks are analysed to identify which spatio-temporal and numerical characteristics are most strongly associated with offensive success. The resulting framework is further applied in a DBU-focused case study in order to investigate how Danish national teams compare with relevant tournament benchmarks in terms of counter-attacking patterns and outcomes.

The central research question of this project is:

How can counter-attacks following a possession gain be defined reproducibly from synchronized event and tracking data, and how can a human-guided machine learning model refine their detection and support actionable analysis of offensive success in elite football?

To address this overall question, the project is structured around the following subquestions:

- How can a transparent and reproducible rule-based definition of counter-attacks be constructed from synchronized event and tracking data?
- How reliable is this definition when compared with human football judgment?
- Can a human-guided machine learning model learn and refine the counter-attack concept from engineered event and tracking features?
- Which spatio-temporal and numerical characteristics of identified counter-attacks are most strongly associated with offensive success?
- What tendencies can be identified in Denmark’s counter-attacking play compared with tournament benchmarks?

2.2 Scope

This study is limited to the analysis of open-play counter-attacking situations following explicit possession gains, primarily interceptions and possession-winning duels. Only transitions occurring within a predefined temporal window after the regain of possession are considered. Set pieces, structured build-up phases, prolonged possession sequences, and defensive transition analysis are outside the scope of the project.

3 Data

3.1 Tracking Data Description

Table 1: Overview of columns in the tracking data

Column name	Description
label	Text label identifying the competing teams (home team and away team).
half	Match period identifier. Values start at 1 (first half) and may extend to later periods (e.g. extra time and penalties) depending on the match.
time_passed_in_half	Time in seconds since the start of the current match period; resets to 0 at the beginning of each period.
total_time_passed	Cumulative match time in seconds from the start of the match; equals <code>time_passed_in_half</code> in the first period and otherwise includes the duration of previous periods.
home_ <i>i</i> _x	Home player x-coordinate in meters for shirt number <i>i</i> . One column exists for each shirt number observed in the match.
home_ <i>i</i> _y	Home player y-coordinate in meters for shirt number <i>i</i> .
away_ <i>j</i> _x	Away player x-coordinate in meters for shirt number <i>j</i> .
away_ <i>j</i> _y	Away player y-coordinate in meters for shirt number <i>j</i> .
ball_x	Ball x-coordinate in meters, expressed in the same coordinate system as player positions.
ball_y	Ball y-coordinate in meters, expressed in the same coordinate system as player positions.

The tracking data used in this project is provided by the Danish Football Association (DBU) and consists of high-frequency positional data for all players and the ball. The dataset covers all matches from the UEFA EURO 2024 (men), UEFA Women’s EURO 2025, and UEFA Under-21 EURO 2025 tournaments. For each match, the tracking data provides a continuous spatio-temporal representation of match play.

The data is sampled at a fixed frequency of 25 Hz for the men’s tournaments and a fixed frequency of 50 Hz for the womens tournament, corresponding to one observation every 0.04 seconds and 0,02 seconds. Each row in the dataset represents a single time frame and contains the positions of all players on the pitch as well as the position of the ball at that specific moment in time.

3.1.1 Temporal Structure

Each observation includes several time-related variables. The variable `half` indicates the current match period, starting at 1 for the first half and increasing for subsequent periods such as extra time and penalty shootouts if applicable. The variable `time_passed_in_half` measures the

elapsed time in seconds since the start of the current half and is reset to zero at the beginning of each new period. In addition, the variable `total_time_passed` provides a cumulative time measure in seconds from the start of the match, summing the durations of all preceding halves. This allows tracking data from different match periods to be aligned on a single continuous timeline.

3.1.2 Coordinate System

All positional data in the tracking dataset is expressed in a pitch-based Cartesian coordinate system measured in meters. The origin of the coordinate system is located at the center of the pitch. The x-axis represents the length of the field and the y-axis represents the width of the field. In the normalized DBU tracking format used in this project, the coordinate ranges are $x \in [-52.5, 52.5]$ and $y \in [-34, 34]$, corresponding to a pitch of dimensions 105×68 meters. In figure 1, a representation of pitch coordinates is illustrated.

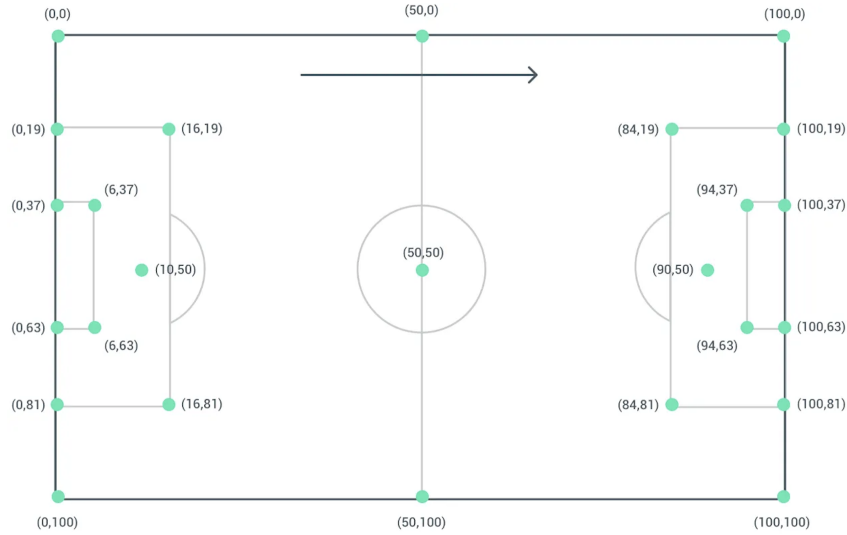


Figure 1: Pitch coordinates

In practice, football pitches may vary slightly in length and width across different stadiums and competitions. These variations are handled by the tracking data providers, who normalize the raw positional data from each stadium to their own internal coordinate systems. The data is subsequently normalized by DBU into a common coordinate system before being delivered for analysis. As a result, all matches in the dataset are provided in a consistent spatial format, enabling direct comparison of positional data across matches, tournaments, and venues.

Occasionally, positional values may slightly exceed the nominal coordinate boundaries. This can occur due to measurement noise, smoothing procedures applied during tracking, or player and ball positions extending marginally beyond the official pitch boundaries, for example during throw-ins or corner situations. Such deviations are typically small and do not affect the overall spatial interpretation of the data.

For the purposes of this project, the normalized coordinate system is treated as fixed and consistent across all matches. Analyses are therefore conducted in the normalized pitch space rather than attempting to recover stadium-specific dimensions. While this approach enables robust cross-match comparisons, it may introduce minor spatial uncertainties, which I consider acceptable given the scale and objectives of the analysis.

3.1.3 Player and Ball Positions

Player positions are provided separately for the home and away teams. For the home team, each player is represented by a pair of columns `home_i_x` and `home_i_y`, where i denotes the player's shirt number. Correspondingly, away team players are represented by `away_j_x` and `away_j_y`. If a player is not present on the pitch at a given time frame, for example due to substitution or dismissal, the corresponding positional values are missing.

The position of the ball is given by the variables `ball_x` and `ball_y`, expressed in the same coordinate system as the players. When available, an additional variable `ball_z` provides the vertical position of the ball. The ball position serves as a central reference point for defining possession, movement direction, and spatial relations between players.

3.2 Event Data Description

Table 2: Overview of columns in the event data (`events.csv`)

Column name	Description
<code>matchId</code>	Unique identifier of the match.
<code>label</code>	Textual label identifying the competing teams.
<code>eventId</code>	Unique identifier of the event.
<code>typePrimary</code>	Primary type of the event (e.g. pass, interception, duel, shot).
<code>matchPeriod</code>	Match period identifier (first half, second half, extra time).
<code>minute</code>	Match minute in which the event occurs.
<code>second</code>	Match second within the given minute.
<code>matchTimestamp</code>	Event timestamp in seconds on the global match timeline.
<code>videoTimestamp</code>	Corresponding timestamp in the broadcast video.
<code>relatedEventId</code>	Identifier linking related events (e.g. duel or pass sequences).
<code>playerId</code>	Identifier of the player performing the event.
<code>teamId</code>	Identifier of the team performing the event.
<code>playerPosition</code>	Nominal playing position of the player.
<code>teamFormation</code>	Formation of the player's team at the time of the event.

Continued on next page

Column name	Description
opponentTeamFormation	Formation of the opposing team at the time of the event.
x, y	Spatial coordinates of the event location on the pitch.
nextX, nextY	Coordinates of the next relevant event location, if applicable.
xG	Expected goals value associated with a shot event.
bodyPart	Body part used to perform the action.
onTarget	Indicator specifying whether a shot is on target.
isGoal	Indicator specifying whether a shot results in a goal.
accuratePass	Indicator specifying whether a pass is successful.
xT	Expected threat value associated with a pass or carry.
successfulCarry	Indicator specifying whether a carry is successful.
carryProgression	Measure of forward progression during a carry.
aerialWon	Indicator specifying whether an aerial duel is won.
groundDuelType	Type of ground duel (e.g. tackle, take-on).
groundDuelKeptPossession	Indicator specifying whether possession is retained after a ground duel.
groundDuelProgressedWithBall	Indicator specifying whether the duel results in forward progression.
groundDuelStoppedProgress	Indicator specifying whether the duel stops opponent progression.
groundDuelRecoveredPossession	Indicator specifying whether possession is recovered in the duel.
groundDuelTakeOn	Indicator specifying whether the duel is a take-on situation.
numberOfEvents	Number of sub-events associated with the event.
duration	Duration of the event in seconds, if applicable.
posStartX, posStartY	Start position of the event in pitch coordinates.
posEndX, posEndY	End position of the event in pitch coordinates.
teamName	Name of the team performing the event.
teamImageUrl	URL reference to the team's image or logo.
infractionType	Type of infraction for foul events.
yellowcard	Indicator specifying whether a yellow card is issued.
redCard	Indicator specifying whether a red card is issued.

In addition to tracking data, the project uses event data provided by DBU for the same set of matches from the UEFA EURO 2024 (men), UEFA Women's EURO 2025, and UEFA Under-21 EURO 2025 tournaments. It contains 31 matches for Women's EURO 2025, 30 matches for UEFA Under-21 EURO 2025 and 52 matches for UEFA EURO 2024 (men), which makes a sum of 113 total matches. The event data consists of a sequence of discrete, annotated match events

describing specific actions performed by players during the game.

Each row in the dataset represents a single event, such as a pass, duel, interception, foul, or shot. Unlike the tracking data, the event data is not sampled at a fixed frequency but is recorded only at moments where an annotated action occurs.

Table 3: Unique event types in `events.csv` and their frequencies

Event Type	Count	Event Type	Count
acceleration	749	interception	8326
clearance	1286	offside	341
corner	1133	own_goal	15
duel	42014	pass	98460
fairplay	70	penalty	36
free_kick	2878	postmatch_penalty	61
game_interruption	7490	postmatch_penalty_faced	55
goal_kick	1831	shot	2669
goalkeeper_exit	355	shot_against	1026
infraction	2659	throw_in	4290
		touch	15257

Among the listed event types, only a subset is directly relevant for the identification and analysis of counter-attacks in this project. In particular, events that indicate a gain of possession, such as interception and certain instances of duel, are used to define potential starting points of counter-attacking sequences. A change in possession can also be inferred from shifts in `teamId` between consecutive events.

3.2.1 Temporal Information

Each event is associated with a timestamp measured in seconds, indicating the exact moment at which the event occurs. The variable `half` specifies the match period in the same manner as in the tracking data, allowing events to be placed on the same match timeline. This shared temporal reference enables precise alignment between discrete events and the high-frequency positional data.

3.2.2 Spatial Information and Attributes

For each event, spatial coordinates (x , y) are provided, representing the location of the action on the pitch. Unlike the tracking data, however, these coordinates are not expressed in a centered metric coordinate system. Instead, the event data uses a normalized event-based pitch representation in which both axes run on a fixed scale from 0 to 100.

As a result, the event and tracking data are not spatially identical in their raw form. In this project, event data is primarily used to identify and classify on-ball actions such as possession

gains, passes, and shots, while the tracking data serves as the primary source for all spatial and metric calculations involving player and ball positions.

4 Definition of Counter Attacks

4.1 Rationale

Counter-attacks are tactically intuitive, but analytically difficult to define in a fully reproducible way. In football analysis, the term usually refers to a transition attack following a regain of possession, but operational definitions vary considerably across studies and providers (Yang et al. 2025; Sarmiento et al. 2018). As a result, a purely verbal definition is not sufficient for systematic extraction from data.

For this reason, this project defines counter-attacks through a transparent rule-based procedure that can be reproduced directly from structured event and tracking data. The aim is not to claim a universally correct football ontology, but rather to construct a practically useful and methodologically defensible definition that captures sequences which, in tactical terms, resemble genuine counter-attacks.

A central challenge in this process is that counter-attacks do not arise from a uniform spatial context. A possession regain close to the team’s own goal and a regain close to the opponent penalty area should not be judged by identical temporal or action-based constraints. This makes a single fixed-threshold definition problematic. The final setup therefore combines event-based regain detection with tracking-based spatial measurements and distance-aware scaling of key thresholds.

4.2 Design Principles

The final definition was built around four principles:

1. **Tactical validity:** the sequence should begin with a genuine regain of possession and reflect a rapid attacking transition toward the opponent goal.
2. **Context sensitivity:** the definition should adapt to where on the pitch the regain occurs, rather than applying the same fixed requirements to all situations.
3. **Reproducibility:** all inclusion criteria should be derived algorithmically from structured data without manual identification of counter-attacking situations.
4. **Analytical usability:** the final sample should be large enough for meaningful downstream analysis, while avoiding clear false positives.

4.3 Development Process

The definition was developed iteratively and changed substantially during the process.

Step 1: Initial fixed-threshold prototype. The first version of the model was based on fixed thresholds for duration, progression, directness, and pass count. This was a natural starting point, partly inspired by previous rule-based approaches to transition detection (Yang et al. 2025). The initial logic was useful as a baseline because it provided a clear and interpretable first extraction of candidate counter-attacks.

However, after inspecting the extracted sequences, it became clear that the fixed-threshold setup had important weaknesses. In particular, the same duration and action limits were being imposed on all regains regardless of whether the regain took place deep in the team’s own half or close to the opponent goal. This created an imbalanced sample, where some deep but tactically plausible counter-attacks were excluded, while some short-distance sequences were retained even though they no longer looked like genuine transition attacks.

Step 2: Visual validation and identification of model weaknesses. To evaluate whether the extracted sequences actually resembled counter-attacks, I generated sequence-overview plots combining player trajectories, ball trajectory, and speed information. These plots were used as a qualitative validation layer, and a subset of borderline sequences was also checked against available video clips.

This validation step revealed three recurring issues:

- **Uneven treatment of long and short regains.** The fixed-threshold version penalized regains far from goal, where longer progression is naturally required, while being too permissive for regains closer to goal.
- **Very short live-ball episodes.** Some extracted sequences had the formal structure of a transition in the event data, but the ball was only in play for a very short time. In extreme cases, the ball was in play for around 0.6 seconds. These situations were often restart-like episodes, such as throw-ins, where the ball and surrounding players moved quickly enough to mimic a transition pattern in raw data, even though the attacking phase never meaningfully developed. An example is shown in Figure 2.
- **Mismatch between event-based and tracking-based spatial information.** The event data was useful for identifying regain events and sequence boundaries, but its raw coordinates were not metrically aligned with the tracking coordinates. In several borderline cases, the regain location inferred from the event data did not fully match the location suggested by the tracking-based ball position and the video. This made it clear that event data should primarily be used for event logic, while tracking data should be used for spatial measurement.

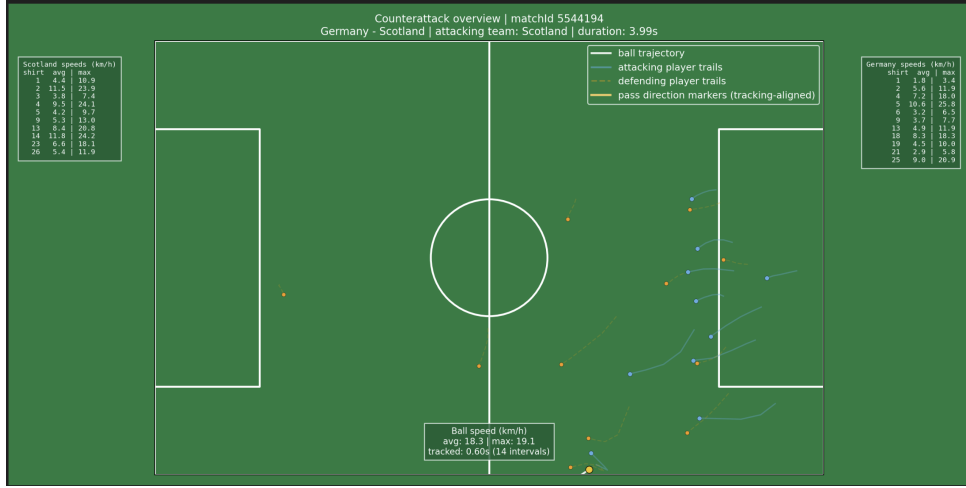


Figure 2: Example of a false positive in an earlier version of the model, where the ball was in play for only 0.6 seconds.

Step 3: Tracking-based and distance-aware recalibration. These observations led to a substantial redesign of the extraction procedure. The final model combines the components summarized in Table 4.

Table 4: Main components introduced in the final extraction setup

Component	Purpose
Event-based regain logic	Identifies candidate transition starts
Tracking-based spatial features	Measures start position and progression
Distance-scaled duration and pass limits	Adapts constraints to start distance from goal
Smooth minimum duration	Filters underdeveloped transitions
Minimum in-play duration	Removes short restart-like episodes
Goal-progress speed requirement	Ensures rapid progression toward goal

This final setup was not chosen in one step. Rather, it emerged through repeated testing, visual inspection, and refinement. In that sense, the final definition is best understood as the result of a calibration process in which tactical plausibility and analytical usefulness were balanced against each other.

4.4 Operational Definition of a Counter-Attack Attempt

A sequence is classified as a counter-attack attempt if all of the following conditions are satisfied.

1) Start condition: possession regain

A sequence must begin with an open-play possession regain: either an **interception** or a possession-winning **duel**. The candidate is accepted only if team possession changes relative to the previous controlled on-ball event.

2) Separation of event logic and spatial measurement

The event data is used to identify the regain, the team in possession, and the sequence stop events. The tracking data is then used to obtain the spatial position of the ball and to measure start location, progression, and speed.

More specifically, the regain time is taken from the event data via `matchTimestamp`, converted to seconds, and aligned to the nearest tracking frame. Spatial features are then computed from the tracking-based ball coordinates rather than from the raw event coordinates. This change was introduced because the event and tracking data do not use the same raw coordinate representation.

3) Sequence stop condition

A counter-attack sequence ends when the first stopping condition occurs: the scaled maximum time budget is exceeded, a `game_interruption` or restart event (`throw_in`, `goal_kick`, corner occurs), or the opponent performs the next controlled on-ball event.

Scaled rules instead of fixed rules

As discussed, the initial fixed-threshold prototype had important weaknesses. Most importantly, it treated all possession regains as if they arose from the same tactical context. This meant that regains far from the opponent goal were constrained by the same duration and pass limits as regains close to goal, even though deeper transitions naturally require more progression time and often involve more actions. Conversely, high regains close to the opponent goal could be allowed to continue for too long under a fixed rule set, even when they no longer resembled genuine counter-attacks. To address these weaknesses, the final definition replaced fixed thresholds with scaled thresholds, so that the rules adapt to the spatial context of the regain.

Final Parameterization

The final rule-based extraction setup is controlled by a set of calibrated parameters governing duration limits, pass limits, territorial constraints, and progression-speed requirements. The parameter values were determined through iterative calibration based on visual sequence inspection, qualitative validation against video examples, and tactical plausibility considerations.

Table 5 summarizes the main parameters used throughout the extraction procedure.

Table 5: Main parameters used in the final rule-based extraction setup.

Parameter	Value	Description
D_{ref}	52.5 m	Reference distance from midfield to the opponent goal
T_{ref}	17.0 s	Reference maximum duration at D_{ref}
P_{ref}	5 passes	Reference maximum pass count at D_{ref}
t_{min}	2.5 s	Absolute minimum allowed sequence duration
$t_{\text{in-play,min}}$	2.5 s	Minimum required live-ball duration
$s_{\text{end,min}}$	52.5 m	Minimum required end position (opponent half)
$v_{\text{min,mid}}$	12.5 km/h	Minimum progression speed near midfield
$v_{\text{min,opp}}$	9.0 km/h	Minimum progression speed near the opponent goal

Here, $D_{\text{ref}} = 52.5$ m corresponds approximately to the distance from midfield to the centre of the opponent goal in the normalized pitch coordinate system. The duration and pass-count limits are scaled relative to this reference distance. Speed thresholds are specified in km/h for interpretability, but are converted to m/s when used in the progression-speed calculations.

4) Distance-scaled maximum duration and pass count

Conceptually, the scaling mechanism allows counter-attacks starting deeper on the pitch to develop over longer durations and with more actions, while applying tighter constraints to high regains closer to the opponent goal. The scaling is therefore intended to adapt the definition to the tactical context of the regain.

Let D_0 denote the tracking-based Euclidean distance from the regain location to the opponent goal centre. Maximum duration and pass count are scaled by the ratio between D_0 and the reference distance D_{ref} :

$$T_{\text{max}} = \max \left(t_{\text{min}}, T_{\text{ref}} \cdot \frac{D_0}{D_{\text{ref}}} \right),$$

$$P_{\text{max}} = \max \left(1, \left\lceil P_{\text{ref}} \cdot \frac{D_0}{D_{\text{ref}}} \right\rceil \right).$$

Thus, deeper regains are allowed more time and more passes, while high regains must develop more quickly and directly.

5) Smooth minimum-duration requirement

Let $s_0 \in [0, 105]$ denote the ball’s forward territorial position at the start of the sequence, measured in meters from the team’s own goal line toward the opponent goal. To filter underdeveloped transitions, the minimum required duration is defined by linear interpolation between three anchor

points: 4.0 s at the own goal line, 3.25 s at midfield, and 2.5 s at the opponent goal line:

$$T_{\min}(s_0) = \begin{cases} 4.0 - \frac{0.75}{52.5}s_0, & 0 \leq s_0 \leq 52.5, \\ 3.25 - \frac{0.75}{52.5}(s_0 - 52.5), & 52.5 < s_0 \leq 105. \end{cases}$$

6) Goal-progress speed requirement

Progression is measured as reduction in distance to the opponent goal. If $D_{\min}$ is the minimum goal distance reached during the sequence, then:

$$\Delta D = D_0 - D_{\min}, \quad v_{\text{prog}} = \frac{\Delta D}{\text{duration}}.$$

where ΔD is measured in meters and duration in seconds, such that v_{prog} is expressed in m/s.

The minimum required progression speed is adjusted according to the spatial start position of the sequence.. It is 12.5 km/h up to midfield and then decreases linearly to 9.0 km/h at the opponent goal line:

$$v_{\min}(s_0) = \begin{cases} 12.5, & s_0 \leq 52.5, \\ 12.5 + \frac{s_0 - 52.5}{52.5}(9.0 - 12.5), & 52.5 < s_0 < 105, \\ 9.0, & s_0 \geq 105. \end{cases}$$

The final required speed is:

$$v_{\text{req}} = \max\left(\frac{D_0}{T_{\max}}, \frac{v_{\min}(s_0)}{3.6}\right),$$

where the division by 3.6 converts the speed threshold from km/h to m/s in order to match the units of v_{prog} . The sequence is accepted only if:

$$v_{\text{prog}} \geq v_{\text{req}}.$$

7) In-play and end-location requirements

Finally, the sequence must contain at least 2.5 seconds of live-ball time and must end in the opponent's half:

$$t_{\text{in-play}} \geq 2.5, \quad s_{\text{end}} \geq 52.5.$$

The first condition removes very short or restart-like episodes, while the second excludes transitions that break down before reaching a sufficiently advanced territorial state.

4.5 Success Labels

After extraction, each attempt is evaluated through event-based outcome labels:

- **Primary success:** at least one shot in the sequence.
- **Secondary outcomes:** box entry and final-third entry.

This preserves a separation between the definition of a counter-attack and the evaluation of whether it leads to a dangerous attacking outcome.

4.6 Why This Setup Is More Reliable

Compared with the original fixed-threshold formulation, the final setup improves reliability in several ways.

- **It handles spatial context more fairly.** By scaling the maximum duration and pass count with start distance to goal, the setup avoids applying identical constraints to fundamentally different tactical situations.
- **It uses tracking data for the quantities where spatial precision matters most.** This reduces dependence on raw event coordinates when computing start position, progression, end position, and speed.
- **It measures progression in tactical rather than purely geometric terms.** Goal-progress speed captures whether the ball is moving effectively toward the opponent goal, rather than simply moving fast in any direction.
- **It applies a context-sensitive speed requirement.** The minimum required speed is no longer fixed across the whole pitch. Instead, it is relaxed progressively in advanced attacking areas, which makes the setup more realistic for sequences starting high up the field, where less absolute progression is needed to create danger.
- **It requires the attack to reach a meaningful territorial outcome.** By requiring the sequence to end on the opponent’s half, the setup excludes transitions that begin like counter-attacks but break down before the attack reaches a sufficiently advanced area.
- **It reduces false positives from undeveloped or restart-like episodes.** This is achieved by combining minimum in-play duration with a minimum sequence-duration requirement.

Taken together, these choices produce a sample that is both tactically plausible and analytically usable.

4.7 Relation to Prior Literature

The final model is literature-informed rather than copied directly from a single source.

- The general rule-based approach to automatic transition detection is aligned with recent work on football counter-attack detection (Yang et al. 2025).

- The emphasis on rapid progression, attacking effectiveness, and the special value of transitions is consistent with earlier empirical studies in elite football (González-Rodenas et al. 2020; Tenga et al. 2010; Sarmento et al. 2018; Michailidis et al. 2024; Eusebio et al. 2025).

The main methodological contribution of the present setup is therefore not a new football-theoretical concept of counter-attacks, but rather the adaptation of existing transition ideas to the specific DBU event and tracking data structure through explicit calibration and validation.

4.8 Limitations

Despite the improvements, several limitations remain.

- **Annotation dependency.** Regain detection still depends on event annotations, especially duel-related outcome fields. Some true regains may therefore be missed.
- **Tracking-event alignment.** Although the regain event is aligned to the nearest tracking frame, event timestamps and tracking timestamps are not perfectly identical. This may affect boundary decisions in short sequences.
- **Rule abstraction.** Even a refined rule-based model cannot capture all tactical intentions, local numerical superiority, or defensive disorganization.
- **Potential survivorship bias.** The final setup includes minimum duration and minimum in-play filters. These are useful for removing cases where an attempted transition immediately breaks down, but they also mean that some early, unsuccessful transition attempts are excluded. Consequently, the extracted sample should be interpreted as a set of counter-attacks that meaningfully develop, rather than every possible initial intention to counter-attack.
- **Dataset specificity.** The thresholds are calibrated to the present DBU dataset and may require adjustment for other competitions, providers, or event annotation systems.

Accordingly, the final definition should be interpreted as a reproducible analytical approximation of counter-attacking play, not as a universally valid football ontology.

4.9 Qualitative Validation

As a qualitative validation step, sequence-overview plots were generated for sampled sequences, combining player trajectories, ball trajectory, and player and ball speed summaries. These plots were used to inspect whether extracted sequences resembled genuine counter-attacks in tactical terms. In addition, a subset of borderline cases was compared against available video clips. This did not create a formal ground-truth annotation set, but it did provide an important face-validity check and directly informed several changes to the final setup, especially the move from event-based spatial features to tracking-based spatial features.

An example of the sequence-overview visualization is shown in Figure 3.

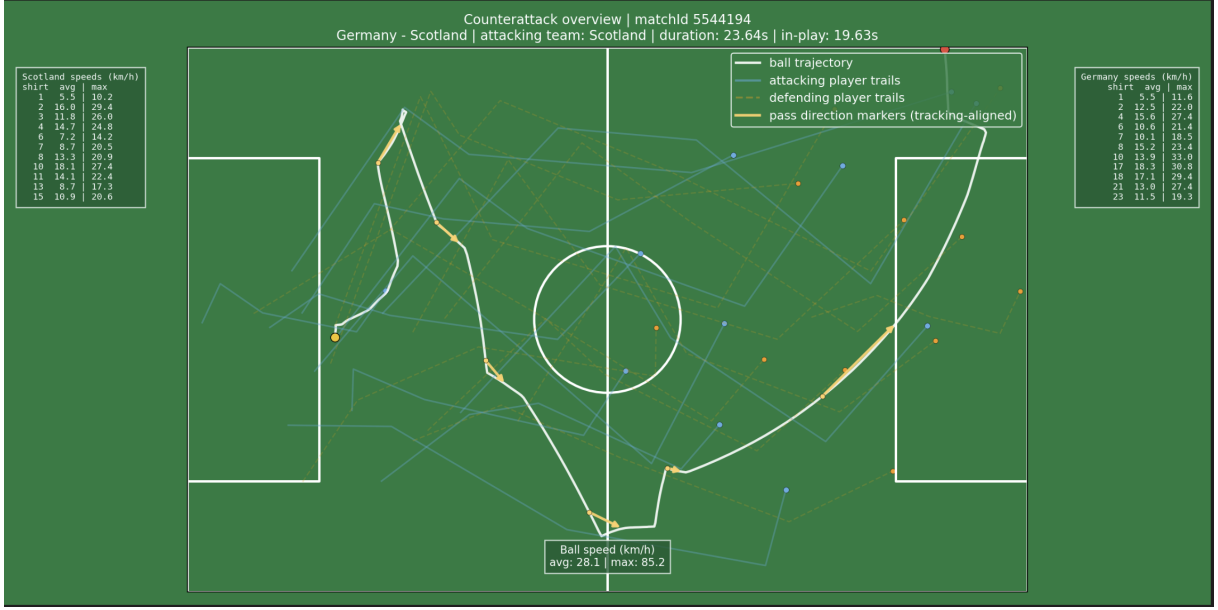


Figure 3: Example of sequence-overview plot used for qualitative validation of extracted counter-attacks.

4.10 Extraction Output and Sanity Checks

Applying the final rule-based definition yielded $N = 746$ counter-attack attempts across the dataset. The extracted sample shows the following outcome profile in Table 6

Table 6: Outcome profile of extracted counter-attacks.

Outcome metric	Rate
Shot rate	22.39%
Shot from box rate	17.29%
Box entry rate	56.84%
Final-third entry rate	81.37%

In terms of sequence characteristics, the median in-play duration is 7.75 seconds, the median progression toward goal is 39.43 meters, and the median average ball progression speed toward goal is 4.57 m/s (≈ 16.45 km/h). The median number of passes is 2. Overall, this is consistent with a fast-transition attacking profile rather than sustained possession play.

The start-zone distribution is:

- **Defensive third:** 327 attempts (43.83%)
- **Middle third:** 280 attempts (37.53%)

- **Attacking third:** 139 attempts (18.63%)

This distribution is tactically plausible. Most counter-attacks still begin in the defensive or middle third, where transition space is typically greatest, but the final setup also retains a meaningful number of high regains in the attacking third when they satisfy the tempo and development criteria. This is desirable, since not all genuine counter-attacks necessarily begin deep.

Overall, these diagnostics suggest that the final extraction setup is operationally stable, tactically credible, and suitable for subsequent analysis of which counter-attacking patterns are associated with offensive success.

4.11 Confusion Matrix Validation

To evaluate the reliability of the final rule-based counter-attack definition, a confusion matrix was constructed by comparing the pipeline output with a human consensus label on a focused validation sample. Since a confusion matrix requires both positive and negative examples, the sample was not created from the final model alone. Instead, an earlier and broader counter-attack definition was used as a candidate generator. This broader setup was intentionally chosen because it captured many plausible attacking transitions, including both genuine counter-attacks and clear false positives. As a result, the validation sample contained sequences that the final definition should accept, sequences that it should reject, and potentially some genuine counter-attacks that the final definition might miss.

The validation sample was created from three matches:

- Spain – Germany MEN’S EURO 2024
- Poland – Sweden WOMAN’S EURO 2025
- Norway – Italy WOMAN’S EURO 2025

For these three matches, video material was obtained and all selected candidate sequences were manually assembled into one review video. The final validation set contained 48 sequences. Each sequence was then evaluated independently by seven raters, all of whom were my personal football teammates with substantial football experience and many years of playing background. Each rater assigned a binary label: counter-attack (1) or not a counter-attack (0). The final human label for each sequence was defined as the median of the seven ratings. Because the labels were binary and the number of raters was odd, this is equivalent to a majority vote.

The pipeline prediction was taken directly from the final rule-based definition, where each sequence was labeled as either counter-attack (1) or not counter-attack (0). The confusion matrix therefore compares the current final pipeline against the aggregated human consensus.

Table 7: Confusion matrix for the final rule-based counter-attack definition against the human consensus label. (CA = Counter-Attack)

	Pipeline: no CA	Pipeline: CA
Human consensus: no CA	17	6
Human consensus: CA	6	19

From Table 7, the rule-based definition produced 19 true positives, 17 true negatives, 6 false positives, and 6 false negatives. This corresponds to the performance measures shown in Table 8.

Table 8: Performance metrics derived from the confusion matrix.

Metric	Precision	Recall	Specificity	Accuracy	F1-score
Value	0.760	0.760	0.739	0.750	0.760

Overall, the results indicate a substantial level of agreement between the final rule-based definition and the human consensus. The definition correctly classified 36 out of 48 sequences, corresponding to an accuracy of 75.0%. The recall of 76.0% shows that the framework captures most of the sequences that the human raters considered to be counter-attacks, while the precision of 76.0% indicates that most sequences identified as counter-attacks by the pipeline were also accepted by the human raters. The specificity of 73.9% further suggests that the framework is reasonably capable of rejecting sequences that do not resemble counter-attacks.

The false positives and false negatives are fully balanced (6 versus 6), indicating that the final definition is neither strongly over-inclusive nor overly conservative. Instead, the disagreements appear primarily in borderline transition situations where tactical interpretation is naturally less clear. This is expected for a football concept such as counter-attacks, which is intuitive in practice but difficult to formalize perfectly from structured data alone.

Taken together, the confusion matrix supports the conclusion that the final rule-based definition is sufficiently reliable for the subsequent analyses. At the same time, the remaining disagreement with human judgment illustrates that the definition should still be interpreted as a reproducible analytical approximation rather than a perfect representation of the underlying tactical concept.

Inspection of disagreement cases

To better understand where the final rule-based definition diverged from human judgment, a small set of disagreement cases was inspected qualitatively using the corresponding sequence snapshots, event annotations, and tracking-derived sequence characteristics. This inspection showed that the disagreements were not random, but instead reflected a small number of recurring

mechanisms related to advanced regains, underdeveloped attacking sequences, and the newly introduced territorial end-location requirement.

A first false-positive case is clip 13 shown in figure 4. (Poland – Sweden, Sweden), which was accepted by the model but rejected by the human raters. Formally, the sequence satisfied all active constraints: it lasted 6.92 seconds, involved two passes, progressed 29.41 meters toward goal, and exceeded the required speed threshold with an average progression speed of 4.25 m/s against a requirement of 3.47 m/s. However, the sequence did not reach the final third or the penalty box and ended shortly after crossing into the opponent’s half. This suggests that the current definition can still classify some rapid regains with immediate forward progression as counter-attacks even when they appear tactically underdeveloped to human observers.

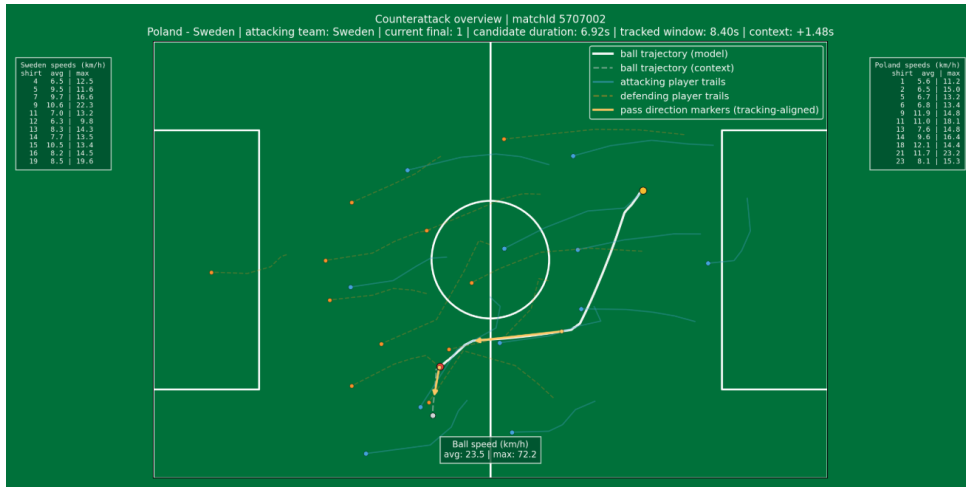


Figure 4: Clip 13, False-Positive

A representative false-negative case is clip 27, shown in figure 5. (Norway – Italy, Norway), which was rejected by the model despite being judged as a counter-attack by all human raters. In this case, the disagreement is best explained by a technical limitation in the earlier broad candidate-generation model used to construct the validation sample. In that broader setup, there was an error in the stop logic such that sequences were terminated as soon as a duel occurred, even in situations where the duel was in fact won and the attack continued. As a result, this sequence was truncated too early and ended before the ball had crossed midfield. The video clip that was distributed to the raters, however, clearly showed that the attack continued beyond this point and effectively extended the situation by approximately three seconds compared with the endpoint shown in Figure 7. Had the sequence been allowed to continue beyond the duel, the final definition would also have classified it as a counter-attack, since the attack then progressed sufficiently far up the pitch. Consequently, this disagreement should not primarily be interpreted as a substantive error in the final definition itself, but rather as a boundary artifact inherited from the earlier broad model used to generate the snapshot sequence on which the validation was based.

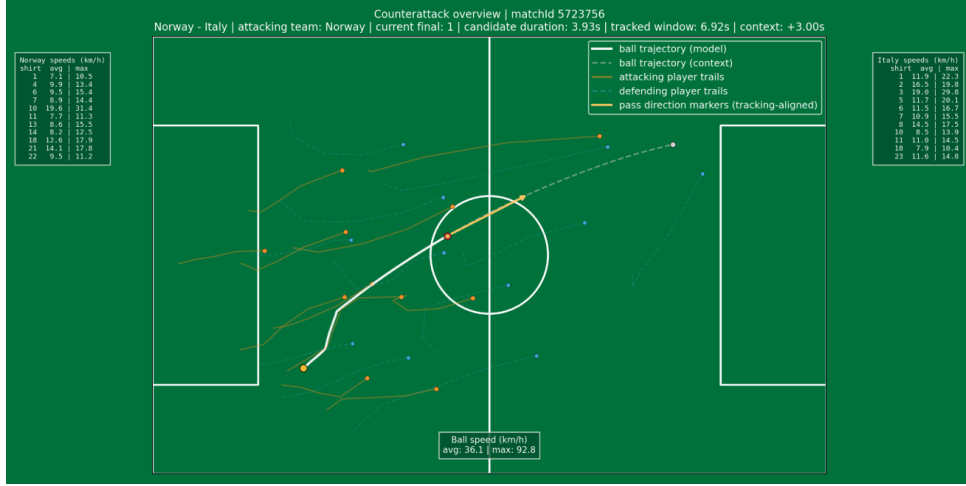


Figure 5: Clip 27, False-Negative

Taken together, these cases suggest that the remaining disagreements between the final rule-based definition and human judgment arise from a combination of modeling choices and validation-sample construction. On the one hand, the final definition can still be somewhat permissive for short or very advanced regain situations that satisfy the formal tempo and progression criteria but are not always perceived by human observers as genuine counter-attacks. On the other hand, the precision estimated from the confusion matrix should be interpreted with some caution, since the validation sample was generated from an earlier broad candidate-definition. As shown by figure 5, this broader setup contained boundary errors in some duel situations, which could truncate sequences too early and thereby alter the snapshot that was later presented for human assessment. Consequently, part of the disagreement between model and human labels may reflect imperfections in the construction of the validation sample rather than only weaknesses in the final definition itself. More generally, this highlights that sequence-based validation remains sensitive to event-tracking alignment, stop-logic implementation, and the exact temporal boundaries used for visual review.

5 Human-Guided Machine Learning Classification

5.1 Machine Learning Model for Human-Guided Counter-Attack Detection

I implemented a machine learning component to the project, to complement the rule-based definition with a data-driven model capable of learning more complex patterns from synchronized event and tracking data. Since the available data are tabular rather than image-based, gradient-boosted decision trees were considered more appropriate than deep vision-based approaches. In particular, XGBoost was chosen because it is a strong and widely used method for structured classification tasks, and because it can model non-linear relationships and interactions between features without requiring these interactions to be specified manually in advance (Chen and Guestrin 2016). This was important in the present setting, where the importance of features

such as progression speed, territorial gain, and event type may depend on each other and on contextual factors such as where on the pitch the regain occurs.

In practical terms, XGBoost constructs an ensemble of decision trees sequentially, where each new tree is trained to reduce the remaining prediction error of the existing ensemble (Chen and Guestrin 2016). This makes the model well suited for the present problem, since counter-attacks are unlikely to be separable by a single linear rule and are better understood as a combination of several interacting spatio-temporal features. The model output is a probability of whether a regain candidate should be classified as a counter-attack, which is then converted into a binary prediction by means of a chosen classification threshold.

5.1.1 Initial Baseline Model Based on the Rule-Based Definition

I started with a machine learning model that was trained directly on labels generated by the final rule-based definition. The purpose of this model was initially not to replace the definition, but to test whether a supervised model could learn to reproduce the counter-attack concept encoded in the definition from a broader candidate pool. The candidate pool contained 10,649 regain situations, of which 745 were labeled as counter-attacks by the rule-based definition and 9,904 as non-counter-attacks. This corresponds to a strongly imbalanced classification problem, with the positive class constituting only about 7% of the total candidate set.

To train the model, I defined a set of input features. In total, I engineered 86 candidate features from event and tracking data. These features described, among other things, regain type, first action type, event counts, duration, in-play time, territorial development, goal-directed progression, minimum distance to goal, and average progression speed over multiple time windows. To reduce model complexity and improve interpretability, an initial full model was trained and its features were ranked by XGBoost gain-based feature importance. In the XGBoost framework, gain quantifies how much a feature contributes to improving the objective function when it is used in tree splits, and higher gain therefore indicates greater importance for the final classifier (XGBoost Developers 2026). A top- k search was then performed over several reduced feature sets. The final 30-feature specification is shown in Table 9. The gain values are taken from the rule-trained XGBoost baseline and are reported after aggregating encoded features back to their original feature names. The prefixes in the feature names refer to the post-regain time window over which the feature was calculated: **short** = 2 seconds, **mid** = 5 seconds, **ten** = 10 seconds, **fifteen** = 15 seconds, and **long** = 20 seconds.

Table 9: Top-30 features used in the shared XGBoost model specification, ranked by XGBoost gain.

Feature	Gain	Description
ten_progression_goal_m	124.13	Reduction in distance to the opponent goal within the first 10 seconds after the regain.
long_avg_progress_speed_mps	103.93	Average speed of goal-directed progression over the long post-regain window.
fifteen_progression_goal_m	94.88	Reduction in distance to the opponent goal within the first 15 seconds after the regain.

Feature	Gain	Description
first_action_type	44.98	Type of the first on-ball action by the regaining team after the possession gain.
fifteen_avg_progress_speed_mps	38.46	Average speed of goal-directed progression within the first 15 seconds.
in_play_20s	32.71	Amount of in-play time captured within the long post-regain window.
long_end_progress_m	31.23	Forward pitch position of the ball at the end of the long post-regain window.
in_play_15s	17.26	Amount of in-play time captured within the first 15 seconds after the regain.
long_duration_s	14.48	Duration of the long candidate sequence, up to the 20-second limit or earlier terminal event.
long_progression_goal_m	13.65	Reduction in distance to the opponent goal over the long post-regain window.
fifteen_duration_s	12.08	Duration of the candidate sequence within the 15-second window.
mid_progression_goal_m	9.40	Reduction in distance to the opponent goal within the first 5 seconds.
fifteen_end_progress_m	8.53	Forward pitch position of the ball at the end of the 15-second window.
stop_reason	8.49	Reason the candidate window ended, for example opponent possession, stoppage, or window end.
long_min_goal_distance_m	8.36	Minimum Euclidean distance from the ball to the opponent goal reached in the long window.
ten_min_goal_distance_m	7.68	Minimum Euclidean distance from the ball to the opponent goal reached within 10 seconds.
event_count_pass_15s	7.56	Number of pass events by the regaining team within the first 15 seconds.
fifteen_min_goal_distance_m	6.62	Minimum Euclidean distance from the ball to the opponent goal reached within 15 seconds.
ten_start_goal_distance_m	6.13	Distance from the ball to the opponent goal at the start of the 10-second window.
long_progression_x_m	5.55	Maximum forward x-axis progression reached over the long post-regain window.
fifteen_start_progress_m	5.51	Forward pitch position of the ball at the start of the 15-second window.
mid_min_goal_distance_m	5.50	Minimum Euclidean distance from the ball to the opponent goal reached within 5 seconds.
event_count_shot_10s	5.42	Number of shot events by the regaining team within the first 10 seconds.
event_count_duel_15s	5.29	Number of duel events involving the regaining team within the first 15 seconds.
ten_start_progress_m	5.00	Forward pitch position of the ball at the start of the 10-second window.
fifteen_progression_x_m	4.96	Maximum forward x-axis progression reached within the first 15 seconds.
short_min_goal_distance_m	4.92	Minimum Euclidean distance from the ball to the opponent goal reached within the first 2 seconds.
long_start_progress_m	4.37	Forward pitch position of the ball at the regain/start of the long window.
mid_start_goal_distance_m	3.82	Distance from the ball to the opponent goal at the start of the 5-second window.

Feature	Gain	Description
<code>event_count_shot_5s</code>	1.12	Number of shot events by the regaining team within the first 5 seconds.

After ranking the engineered features by gain, several top- k feature sets were evaluated. The final rule-trained baseline was fixed at the top 30 features, as this specification achieved the highest F1-score among the tested top- k configurations. Under this 30-feature setup, the rule-trained baseline achieved a precision of 0.837, a recall of 0.941, and an F1-score of 0.886 on the internal hold-out split constructed from the rule-labeled candidate pool.

This initial result showed that the model could reproduce the rule-based definition well. However, it also made clear that the model primarily learned the logic of the definition itself rather than an independently human football understanding of counter-attacks. In other words, the model was able to approximate the definition, but this alone did not guarantee that it captured counter-attacks in a way that better reflected human football judgment.

5.1.2 Motivation for a More Human-Guided Model

A central methodological concern that emerged during model development was that a machine learning model trained only on rule-generated labels would remain constrained by the assumptions and limitations of the rule-based detector. If the broader purpose of the machine learning model was to identify counter-attacks in a way that more closely reflected human football judgment, the model needed access to explicitly human labels.

At the same time, a fully manual labeling of the complete regain candidate pool was not realistic. The pool was large, the class distribution was highly imbalanced, and the generation of video clips and manual ratings was time-consuming. This motivated a more selective annotation strategy inspired by active learning under class imbalance, where limited annotation effort is concentrated on the most informative observations rather than distributed randomly across the full candidate set (Ertekin et al. 2007; Zhan et al. 2021).

The resulting annotation design therefore focused on a strategically selected subset of candidates rather than the full dataset. In total, 180 video clips were annotated by seven human raters, sampled from the groups shown in Table 10.

Table 10: Sampling groups for human annotation

Group	Neg. below	Pos. above	Clear neg.	Random neg.	Clear pos.	Random pos.
Clips	50	50	20	20	20	20

This sampling strategy served two purposes. First, the near-threshold cases were expected to be the most informative because they were most likely to reveal false positives and false negatives relative to human judgment. Second, the clear and random cases ensured that the manual labels

were not based only on ambiguous borderline situations. Of the 180 clips, 10 were excluded because of clip-generation issues such as tracking errors or invalid sequence visualization, leaving 170 usable manually annotated cases.

5.1.3 Aggregation of Human Labels and Inter-Rater Agreement

Each of the 170 usable candidates was annotated independently by seven raters. A human majority label was then assigned by simple majority vote. Thus, a candidate was labeled as a human counter-attack if at least four of the seven raters classified it as such. This majority-vote label was used as the human supervision signal in the subsequent hybrid model (Bucarelli et al. 2023).

To evaluate the internal consistency of the human annotation process, inter-rater agreement was assessed using both raw agreement summaries and Fleiss’ kappa. Fleiss’ kappa is specifically designed for the case where more than two raters assign nominal labels to the same set of items and adjusts observed agreement for agreement expected by chance (Fleiss 1971). In addition, raw agreement shares were reported because they provide a direct and intuitive description of how frequently the raters agreed (McHugh 2012).

For each candidate, agreement was summarized in three ways: (1) the share of raters agreeing with the majority label, (2) the share of agreeing rater-pairs, and (3) Fleiss’ kappa, which adjusts observed agreement for chance.

For the older 48-case validation set, the seven raters produced a mean agreement share of 0.899, a mean pairwise agreement of 0.837, and a Fleiss’ kappa of 0.674. Here, the mean agreement share refers to the average proportion of raters agreeing with the majority label for a given candidate, while the mean pairwise agreement measures the average proportion of agreeing rater-pairs across candidates. Fleiss’ kappa captures the same overall consistency while adjusting for agreement expected by chance.

For the larger 170-case annotation set, agreement was even stronger. The unanimous-share, defined as the proportion of candidates for which all seven raters assigned the same label, was 0.771. The mean agreement share was 0.945, meaning that on average 94.5% of raters agreed with the majority label. The mean pairwise agreement was 0.908, indicating that about 90.8% of all possible rater-pairs agreed on a given candidate. Finally, Fleiss’ kappa was 0.790, showing strong chance-corrected agreement across the full annotation set.

Taken together, these results suggest that the human labels were sufficiently consistent to serve as a meaningful source of supervision in the machine learning pipeline. High agreement is generally interpreted as an indication that the labeling task is well-defined and that the resulting annotations can be treated as a reliable supervision signal (OpenTrain AI 2024)

5.1.4 Construction of the Hybrid Human-Guided Model

The next step was therefore not to discard the rule-labeled data, but to combine it with the human annotations in a hybrid learning framework. This was considered more realistic than training a purely human-labeled model from scratch, since 170 human-labeled cases would still be a very limited training set relative to the size and imbalance of the underlying candidate pool.

The hybrid model was constructed as follows. For candidates that had been manually annotated, the rule-based label was replaced by the human majority label. For all other candidates, the original rule-based label was retained. In this way, the final training target was defined by

$$y_i^{\text{hybrid}} = \begin{cases} y_i^{\text{human}}, & \text{if candidate } i \text{ had human annotation,} \\ y_i^{\text{rule}}, & \text{otherwise.} \end{cases}$$

In addition, the manually annotated candidates were upweighted according to how strong the inter-rater agreement was. The rationale was that highly unanimous human judgments should influence the fitted decision boundary more strongly than ambiguous cases. This weighting strategy is closely related to cost-sensitive learning, where individual observations are assigned different importance during training. This approach was used due to the imbalanced classification setting to counteract bias towards the majority class and to emphasize more informative or reliable samples.

The final weighting scheme was defined as $w = 3$ for 4/7 agreement, $w = 5$ for 5/7, $w = 8$ for 6/7, and $w = 12$ for unanimous 7/7 agreement.

This weighting was applied symmetrically for both positive and negative human majority labels. Thus, the model was not encouraged only toward human positives, but toward high-confidence human judgments in general.

5.1.5 Fair Baseline–Hybrid Comparison

A key methodological decision in the final comparison was that the baseline XGBoost model and the hybrid XGBoost model should be evaluated under the same model architecture, the same feature set, and the same training procedure. Otherwise, any observed difference in performance could be driven either by the supervision signal or by differences in model specification, making the comparison difficult to interpret.

For this reason, the shared feature set was not re-optimized separately for the hybrid setting. Instead, both models were trained using the same 30-feature specification that had previously been selected in the baseline development stage, where it yielded the strongest performance among the tested feature configurations. This feature set combined progression features, end-position features, event-count features, duration features, and contextual event descriptors such as stop reason and first action type.

The baseline and hybrid models were then both retrained using this same 30-feature specification, with identical XGBoost hyperparameters and identical preprocessing. The only substantive

difference between the two models was therefore the supervision signal: the baseline model used only rule-based labels, whereas the hybrid model used human overrides and human-based sample weights where available. This made it possible to attribute any difference in performance primarily to the inclusion of human knowledge rather than to a different feature configuration.

5.1.6 Evaluation Metrics

Because counter-attack detection is a strongly imbalanced classification problem, overall accuracy was not treated as the main model selection criterion. With a large majority of negative candidates, a model can achieve apparently high accuracy simply by predicting most cases as non-counter-attacks. For this reason, the evaluation focused primarily on precision, recall, and the F1-score. Precision measures how many predicted counter-attacks are actually relevant, whereas recall measures how many of the relevant counter-attacks are successfully retrieved. The F1-score was used as the main summary measure because it balances the trade-off between the two. This is consistent with the literature showing that precision–recall based evaluation is more informative than accuracy when the positive class is rare (Saito and Rehmsmeier 2015).

5.1.7 Results

The final human-facing comparison was carried out on the same 48 manually annotated candidates that were previously used to validate the rule-based definition of counter-attacks. For this evaluation, the rule-based definition, the shared-feature baseline XGBoost model, and the shared-feature hybrid XGBoost model were all compared against the majority-vote human label.

Table 11: Comparison of the rule-based detector, baseline XGBoost model, and hybrid human-guided XGBoost model against the human-labeled evaluation set.

Model	TP	FP	FN	TN	Precision	Recall	Accuracy	F1-score
Rule-based definition	19	6	6	17	0.760	0.760	0.750	0.760
Baseline XGBoost	21	7	4	16	0.750	0.840	0.771	0.792
Hybrid XGBoost	22	7	3	16	0.759	0.880	0.792	0.815

Table 11 shows that the rule-based definition already achieved a reasonably strong correspondence with the human-labeled evaluation set, with balanced precision and recall and an F1-score of 0.760. The shared-feature baseline XGBoost model improved on this result primarily by increasing recall from 0.760 to 0.840, indicating that the model was able to combine the available features more flexibly than the rule-based detector alone. The hybrid human-guided XGBoost model performed best overall, achieving the highest recall (0.880) and the highest F1-score (0.815), while maintaining precision at approximately the same level as the baseline model.

At the same time, these results should be interpreted with caution. The evaluation is based on a very small annotated test set of only 48 candidates drawn from a much larger pool of regain situations. This means that the reported metrics are sensitive to the exact composition of the

sample and cannot be assumed to provide a fully stable estimate of out-of-sample performance. In other words, the comparison is informative, but still limited in statistical robustness and external validity.

A larger human-annotated evaluation set would therefore be desirable. In particular, an annotation sample on the order of several hundred candidates, would possibly provide a more reliable basis for validating both the rule-based and hybrid models. However, producing such a dataset was beyond the practical scope of the present project, as manual annotation at that scale would have been highly time-consuming. Future work should therefore prioritize the construction of a substantially larger annotated sample in order to validate the hybrid approach more rigorously.

5.1.8 Human-Label Split Perturbation Analysis

The comparison in Table 11 provided an initial indication that the hybrid XGBoost model aligned better with human judgment than both the rule-based detector and the rule-trained baseline model. However, this evaluation was based on a fixed set of only 48 human-labeled candidates. With such a small test set, the estimated precision, recall, and F1-score may be sensitive to exactly which human-labeled candidates happen to be included in the test sample, as performance estimates based on small hold-out sets are known to exhibit high variance and instability (Raschka 2018). To assess whether the observed improvement was stable or merely a consequence of this particular split, I therefore performed a human-label split perturbation analysis.

The purpose of this analysis was to repeatedly perturb which human-labeled cases were used for training and which were held out for testing. The 170 newly annotated cases and the 48 older validation cases gave 218 human-labeled rows in total. Since three candidates appeared in both annotation sets, these were deduplicated, leaving 215 unique human-labeled candidates. In each repetition, a stratified split was created with 48 human-labeled candidates held out as a test set and the remaining 167 unique human-labeled candidates used as the human-guided training subset. The stratification ensured that the class balance between human counter-attacks and non-counter-attacks was kept approximately stable across splits.

For each of the 100 repetitions, the same procedure was followed. First, all human-test candidates were removed from the training pool to avoid leakage. Second, the baseline XGBoost model was trained using only the rule-based labels. Third, the hybrid XGBoost model was trained using the same features, hyperparameters, and preprocessing, but with human majority labels replacing rule labels for the human-train cases. These human-labeled training cases were again weighted according to annotator agreement, with weights of 3, 5, 8, and 12 for 4/7, 5/7, 6/7, and 7/7 agreement respectively. Finally, both models were evaluated on the 48 held-out human-labeled candidates in that split. This produced 100 estimates of precision, recall, F1-score, and accuracy for each model.

Table 12: Split perturbation results across 100 stratified human-label splits. Values report the mean and standard deviation across repetitions.

Model	Precision	Recall	Accuracy	F1-score
Rule-based definition	0.666 ± 0.073	0.725 ± 0.093	0.771 ± 0.052	0.691 ± 0.069
Baseline XGBoost	0.632 ± 0.080	0.727 ± 0.098	0.750 ± 0.057	0.673 ± 0.072
Hybrid XGBoost	0.684 ± 0.088	0.821 ± 0.094	0.797 ± 0.060	0.742 ± 0.072

Table 12 shows that the hybrid model achieved the strongest average performance across the perturbed human-label splits. Its mean F1-score was 0.742, compared with 0.673 for the baseline XGBoost model and 0.691 for the rule-based definition. The improvement was primarily driven by recall, where the hybrid model achieved a mean recall of 0.821, compared with 0.727 for the baseline model and 0.725 for the rule-based definition. This suggests that the human-guided model was better at retrieving candidates that human raters considered counter-attacks, while also maintaining the highest mean precision among the three approaches.

To make the comparison fairer, the models were also compared within each split by calculating the paired difference in performance between the hybrid model and the two alternatives. This avoids comparing models across different test samples and instead asks whether the hybrid model tends to outperform the alternatives on the same held-out human cases.

For each split i , the paired F1 difference was defined as

$$d_i = F1_{\text{hybrid},i} - F1_{\text{comparison},i}.$$

The observed test statistic was the mean paired difference across the 100 splits:

$$T_{\text{obs}} = \frac{1}{100} \sum_{i=1}^{100} d_i.$$

Under the null hypothesis of no systematic difference between the two models, the sign of each paired difference is exchangeable. This corresponds to a paired permutation (sign-flip) test for matched observations. A null distribution was therefore generated by randomly flipping the sign of each paired difference $s_i \in \{-1, 1\}$ and recalculating

$$T^{(b)} = \frac{1}{100} \sum_{i=1}^{100} s_i d_i$$

for $B = 100,000$ Monte Carlo permutations. The one-sided p-value was then estimated as

$$p = \frac{1 + \sum_{b=1}^B \mathbb{I}(T^{(b)} \geq T_{\text{obs}})}{B + 1}.$$

This p-value therefore measures how often a mean improvement at least as large as the observed one would occur if the direction of the paired split-level differences were random.

Table 13: Paired performance differences across 100 split perturbations. Positive values indicate better hybrid performance. The p-value is based on a paired sign-flip permutation test with 100,000 Monte Carlo permutations.

Comparison	Mean $\Delta F1$	Median $\Delta F1$	Improved splits	p -value
Hybrid vs. Baseline XGBoost	+0.070	+0.069	85%	< 0.0001
Hybrid vs. Rule-based definition	+0.051	+0.066	71%	< 0.0001

In addition to the tabular comparison, the distribution of paired F1-score differences was inspected visually. For each split, the difference

$$\Delta F1 = F1_{\text{hybrid}} - F1_{\text{baseline}}$$

was calculated. The distribution is clearly shifted to the right of zero, indicating that the hybrid model achieved higher F1-scores than the baseline in most splits, rather than the average improvement being driven by a small number of favorable cases.

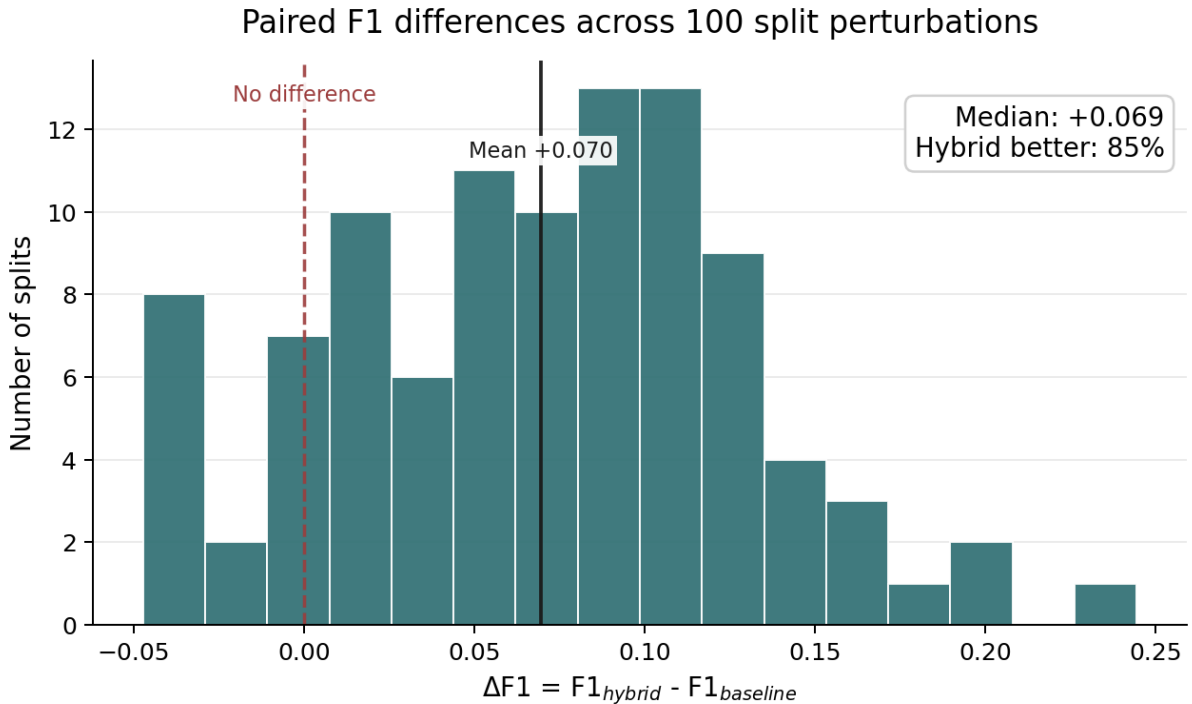


Figure 6: Distribution of paired F1-score differences between the hybrid XGBoost model and the baseline XGBoost model across 100 split perturbations. The vertical dashed line marks zero. Values to the right indicate splits where the hybrid model achieved a higher F1-score.

The paired results in Table 13 and Figure 6 support the same conclusion. The hybrid model improved on the baseline XGBoost model by an average of 0.070 F1-points and achieved a higher F1-score in 85% of the 100 splits. The paired sign-flip permutation test indicated that this

improvement was highly unlikely under the null hypothesis of no systematic difference between the two models ($p < 0.0001$). Compared with the rule-based definition, the hybrid model improved F1 by an average of 0.051 and performed better in 71% of the splits, again with a permutation p-value below 0.0001. This indicates that the improvement from human-guided training was not limited to one favourable fixed test set, but was generally present across many alternative train-test allocations of the available human-labeled data.

At the same time, the analysis also shows that the result is not perfectly stable. The standard deviation of the hybrid model’s F1-score was 0.072, and individual split-level F1-scores ranged from 0.563 to 0.909. This variation is expected given the limited number of human-labeled cases and confirms that performance estimates based on small human test sets should be interpreted cautiously. The split perturbation analysis therefore strengthens the evidence for the hybrid model, but it does not remove the need for a larger held-out human-labeled evaluation set in future work.

Overall, the perturbation analysis provides a more robust basis for interpreting the model comparison. The fixed 48-case evaluation suggested that the hybrid model was best, and the repeated split analysis showed that this pattern was generally maintained across many alternative human-label splits. Within the constraints of the available annotation budget, this supports the conclusion that incorporating human majority labels and agreement-based weights improved the model’s alignment with human football judgment.

5.1.9 Interpretation and Limitations

The model development process therefore moved through three conceptually distinct stages. First, the machine learning model was used as a data-driven approximation of the rule-based definition. Second, human labels were collected in a targeted way to identify where the model and the definition were most likely to disagree with actual football judgment. Third, a hybrid training strategy was introduced in order to incorporate human knowledge without losing the scale advantage of the rule-labeled candidate pool.

The final hybrid model should not be interpreted as a purely human-trained detector, but rather as a human-guided refinement of the original rule-based system. This is an important distinction. The model still depends heavily on the original definition because the vast majority of training examples remain rule-labeled. At the same time, the manually annotated candidates provide targeted corrections in cases where the rule-based definition is most likely to disagree with human judgment, in particular near the decision boundary.

A final methodological limitation concerns the size of the human-labeled evaluation set. The final comparison against human football judgment is based on only 48 annotated candidates drawn from a much larger pool of regain situations. Although the shared top- k feature specification was selected during baseline model development and retained for the hybrid comparison in order to ensure a fair comparison, the resulting human-facing performance estimates should still be interpreted with caution. With such a small annotated evaluation set, the reported

precision, recall, and F1-score are sensitive to the exact composition of the sample and should therefore be viewed as exploratory rather than as a stable estimate of out-of-sample human-aligned performance.

A larger and fully held-out human-labeled test set would be required to support a stronger claim about generalization to human football judgment. In particular, an evaluation sample on the order of several hundred candidates would provide a more robust basis for comparison. However, constructing such a dataset was beyond the practical scope of the present project, as manual annotation at that scale would have been highly time-consuming. Nevertheless, within the available data constraints, the present results still provide consistent evidence that the hybrid human-guided model aligns better with human football judgment than both the rule-based definition and the baseline XGBoost approximation of that definition.

6 Counter-attack Analysis for DBU

6.1 DBU-Focused Counter-Attack Analysis

After developing the hybrid XGBoost model, I used it to produce a more applied analysis aimed specifically at DBU. The purpose of this part of the project was to investigate whether the model output could be translated into practical football insight. I therefore analysed Denmark’s counter-attacks across the men’s senior, women’s senior, and men’s U21 datasets and compared them with relevant benchmarks from the same competitions.

For each dataset, Denmark was compared with three reference groups: the tournament average, the teams reaching the semi-finals/final, and match winners. The tournament average provides a broad baseline, the semi-finalists/finalists represent high-performing teams, and match winners indicate what counter-attacking profiles look like in successful match contexts.

Table 14: Hybrid-detected counter-attacks for Denmark compared with tournament benchmarks.

Dataset	Benchmark	Games	CAs	CA/game	Shot	Box entry	Prog. (m)
Men	Denmark	4	10	2.50	10.0%	40.0%	35.3
Men	Tournament avg.	102	333	3.26	18.3%	55.6%	43.4
Men	Semi-finalists	26	86	3.31	27.9%	66.3%	43.2
Men	Match winners	34	138	4.06	23.9%	62.3%	42.5
Women	Denmark	3	20	6.67	30.0%	60.0%	42.3
Women	Tournament avg.	62	301	4.85	26.2%	58.1%	40.2
Women	Semi-finalists	22	111	5.05	27.0%	55.9%	37.3
Women	Match winners	26	148	5.69	25.0%	64.2%	36.6
U21	Denmark	4	17	4.25	29.4%	47.1%	36.5
U21	Tournament avg.	62	138	2.23	19.6%	50.7%	39.5
U21	Semi-finalists	22	46	2.09	21.7%	58.7%	38.9
U21	Match winners	26	69	2.65	21.7%	53.6%	38.3

The benchmark comparison reveals substantial differences across the three Danish datasets. The

men’s senior team produced only 2.50 hybrid-detected counter-attacks per team-game, below both the tournament average (3.26) and the match-winner benchmark (4.06). Denmark’s attacking outcomes were also substantially weaker than the strongest benchmark groups. The shot rate reached only 10.0%, compared with 27.9% for the semi-finalists, while the box-entry rate was similarly lower at 40.0% compared with 66.3%. Denmark’s counter-attacks also progressed substantially shorter distances toward goal on average (35.3 m) than all benchmark groups, which were above 42 m. Taken together, this suggests that the Danish men’s team both produced fewer counter-attacks and struggled to carry them into dangerous attacking zones.

The women’s senior team showed a markedly different profile. Denmark produced 6.67 counter-attacks per team-game, above both the tournament average and the semi-finalist benchmark. Denmark also achieved the highest shot rate among all benchmark groups at 30.0%. Their progression toward goal was likewise strong, averaging 42.3 metres, which exceeded all benchmark groups. Overall, the women’s dataset therefore suggests a comparatively effective counter-attacking profile.

The U21 team also displayed a high-volume counter-attacking profile. Denmark produced 4.25 counter-attacks per team-game, substantially above the tournament average of 2.23. Their shot rate of 29.4% was likewise higher than all benchmark groups. However, despite the strong shot production, Denmark’s progression toward goal remained lower than the benchmark groups, indicating that many attacks ended relatively early or from less advanced areas.

To investigate the quality of these counter-attacks more directly, I constructed an outcome-based feature profile. The purpose was not only to measure how often Denmark produced counter-attacks, but also to analyse where these attacks ended and how closely they resembled the profiles associated with shot-producing counter-attacks across the tournaments.

This analysis was based on the same post-regain time frame used during feature engineering for the machine learning model. Each regain candidate was described using information from a fixed temporal window after the possession gain, with the longest window covering the first 20 seconds. This window was chosen to capture the immediate transition phase while avoiding that later, more established possession play was incorporated into the counter-attack definition.

For the DBU analysis, I therefore defined an outcome position within this same 20-second window. If the counter-attack produced a shot within the window, the outcome position was defined as the shot location. If possession was lost or the game stopped before a shot occurred, the outcome position was instead defined as the location of that terminal event. If no clear terminal event occurred within the 20-second window, the final tracked ball position inside the window was used. This ensured that all detected counter-attacks could be included in the analysis while remaining aligned with the same temporal structure used during model development.

Based on this outcome definition, I calculated several additional features describing territorial progression, attack speed, timing, directness, and attacking structure. These features made it possible to compare whether Denmark’s counter-attacks ended in similarly dangerous areas as

the counter-attacks that produced shots across the tournaments.

Table 15: Feature profile of hybrid-detected counter-attacks in the men’s dataset.

Feature	Shot	Non-shot	Denmark	Shot minus Non-shot	Denmark minus Shot
Progression toward goal (m)	45.77	42.81	35.31	2.96	-10.47
Outcome progress (m)	88.82	80.52	75.83	8.30	-12.99
Outcome distance to goal (m)	19.49	30.30	34.74	-10.81	15.26
Goal progression speed (m/s)	3.04	3.31	3.38	-0.26	0.33
Duration to outcome (s)	10.41	11.27	8.46	-0.86	-1.95
Directness to goal	0.79	0.85	0.84	-0.07	0.05
Time to first action (s)	1.95	2.32	2.42	-0.37	0.47
Pass count within 20s	1.92	2.42	2.40	-0.50	0.48
Touch count within 20s	0.49	0.43	0.30	0.06	-0.19
Start progression (m)	56.37	42.50	44.99	13.87	-11.38
Start goal distance (m)	52.07	66.52	63.96	-14.45	11.89

The men’s feature profile provides a clearer explanation of Denmark’s weak attacking outcomes. Shot-producing counter-attacks in the tournament progressed substantially further toward goal and ended much closer to the opponent’s goal than Denmark’s attacks. Denmark’s average outcome distance to goal was 34.74 metres, compared with only 19.49 metres for shot-producing counter-attacks. Similarly, Denmark’s average outcome progress reached only 75.83 metres, compared with 88.82 metres for shot-producing attacks.

Importantly, Denmark’s counter-attacks were not slower than successful attacks. In fact, Denmark reached their outcomes faster on average, with a duration to outcome of 8.46 seconds compared with 10.41 seconds for shot-producing counter-attacks. Their progression speed was also slightly higher. The problem therefore does not appear to be tempo itself. Instead, the attacks often ended too early and too far from goal, suggesting insufficient territorial penetration and limited access to dangerous attacking spaces.

Table 16: Feature profile of hybrid-detected counter-attacks in the women’s dataset.

Feature	Shot	Non-shot	Denmark	Shot minus Non-shot	Denmark minus Shot
Progression toward goal (m)	44.53	38.67	42.27	5.86	-2.26
Outcome progress (m)	88.40	84.03	82.90	4.37	-5.50
Outcome distance to goal (m)	19.43	26.79	24.95	-7.36	5.52
Goal progression speed (m/s)	3.26	3.78	4.12	-0.52	0.85
Duration to outcome (s)	8.83	10.56	9.76	-1.74	0.94
Directness to goal	0.88	0.90	0.92	-0.02	0.03
Time to first action (s)	2.09	2.30	2.41	-0.21	0.32
Pass count within 20s	1.56	2.14	1.90	-0.59	0.34
Touch count within 20s	0.56	0.44	0.55	0.12	-0.01
Start progression (m)	58.25	43.87	43.20	14.39	-15.05
Start goal distance (m)	50.65	65.34	63.51	-14.69	12.86

The women’s feature profile is substantially stronger from a Danish perspective. Denmark’s progression toward goal and overall outcome progression were both relatively close to the shot-producing benchmark profile, and the outcome distance to goal was substantially better than the

corresponding men’s values. Denmark’s women’s team also displayed very aggressive attacking behaviour, with the highest progression speed among all profiles and a highly direct attacking structure.

Although Denmark’s attacks still ended somewhat further from goal than the average shot-producing counter-attack, the gap was comparatively moderate. Combined with the strong shot rate and high counter-attacking volume, the results do not indicate a major counter-attacking weakness for the women’s team.

Table 17: Feature profile of hybrid-detected counter-attacks in the U21 dataset.

Feature	Shot	Non-shot	Denmark	Shot minus Non-shot	Denmark minus Shot
Progression toward goal (m)	42.02	38.84	36.45	3.18	-5.57
Outcome progress (m)	88.94	76.19	80.60	12.74	-8.34
Outcome distance to goal (m)	18.31	34.21	29.30	-15.90	10.99
Goal progression speed (m/s)	3.81	3.34	2.72	0.47	-1.08
Duration to outcome (s)	9.33	10.16	10.99	-0.83	1.65
Directness to goal	0.83	0.88	0.79	-0.06	-0.03
Time to first action (s)	1.98	2.13	2.23	-0.15	0.25
Pass count within 20s	1.78	2.05	2.06	-0.27	0.28
Touch count within 20s	0.70	0.45	0.47	0.25	-0.23
Start progression (m)	55.00	43.64	48.90	11.36	-6.10
Start goal distance (m)	54.33	65.74	60.05	-11.41	5.71

The U21 profile presents a more mixed picture. Denmark produced many counter-attacks and achieved a strong shot rate, but the underlying feature profile still differed noticeably from the tournament’s shot-producing benchmark profile. Denmark’s attacks progressed less toward goal, ended further away from goal, and developed more slowly than successful shot-producing counter-attacks.

In particular, Denmark’s outcome goal progression speed was substantially lower than the shot-producing benchmark profile (2.72 m/s versus 3.81 m/s), while the average duration to outcome was longer. This suggests that the U21 team often reached attacking outcomes through slower or less direct transitions. Although the team generated shots relatively frequently, the underlying progression profile indicates that many attacks may not have reached the same level of territorial dominance or defensive imbalance as the strongest tournament counter-attacks.

Overall, the DBU-focused analysis demonstrates that the same hybrid detection framework can reveal very different counter-attacking profiles across teams. For the men’s senior team, the main limitation appears to be insufficient territorial penetration and weak final outcome quality. The women’s senior team displayed the strongest overall transition profile, combining high counter-attacking volume with efficient attacking outcomes. The U21 team produced many counter-attacks and shots, but the underlying progression profile suggests slower and less penetrative attacks than the strongest benchmark sequences.

The analysis should nevertheless be interpreted cautiously. The Danish samples are relatively small, especially for the men’s and women’s senior teams, and the hybrid model detects counter-

attacking situations rather than providing a perfect tactical truth. In addition, the outcome-based feature profile depends on a simplified event-based definition of when the attack reaches its relevant outcome. Despite these limitations, the analysis demonstrates how the hybrid framework can be used as an applied football-analysis tool, not only to count counter-attacks, but also to compare how attacks develop spatially and temporally, where they end, and how closely they resemble successful transition profiles from elite tournament football.

7 Conclusion

This project addressed the challenge of defining and analysing counter-attacks in a reproducible and data-driven way using synchronized event and tracking data. The main objective was to operationalize counter-attacks from structured data and identify which spatio-temporal characteristics are associated with offensive success.

A rule-based definition was developed by combining event-based regain detection with tracking-based spatial measurements and distance-aware thresholds for duration, pass count, and speed. This produced a reproducible extraction procedure that was more context-sensitive than a fixed-threshold approach and generated a tactically plausible sample of counter-attacking situations.

The rule-based definition was validated against human football judgment through manually annotated sequences. The results showed reasonable agreement with the human labels, while disagreement cases mainly occurred in borderline situations such as very short, very advanced, or underdeveloped transitions. This supports the interpretation of the definition as a useful analytical approximation rather than a perfect representation of the tactical concept.

To complement the rule-based setup, an XGBoost model was trained to detect counter-attacks from engineered event and tracking features. A baseline model showed that the rule-based definition could be learned from data, while a hybrid human-guided model incorporated majority-vote human labels and agreement-based sample weights. This hybrid model achieved stronger agreement with the available human-labeled data than both the rule-based detector and the rule-trained baseline model.

Because the human-labeled evaluation set was small, a human-label split perturbation analysis was conducted. Across repeated alternative train-test splits, the hybrid model generally maintained stronger F1-performance, suggesting that the improvement was not driven only by one favourable fixed test set. However, the variation across splits also confirmed that larger held-out human-labeled datasets would be needed for stronger claims about generalization.

Finally, the hybrid model was applied in a DBU-focused analysis. For the Danish men’s senior team, counter-attacking volume was not dramatically below tournament benchmarks, but the attacks were less often converted into dangerous outcomes and tended to end farther from goal. The women’s and U21 teams showed stronger counter-attacking profiles, although the U21 team still showed some gap in final outcome quality.

Overall, the project demonstrates that counter-attacks can be extracted, validated, and analysed reproducibly using combined event and tracking data. The main contribution is a practical framework that combines rule-based logic, human annotation, and machine learning to produce interpretable counter-attack analysis. The framework remains limited by small human-labeled samples, dependence on event annotations, and the fact that observable data cannot fully capture tactical intention. Nevertheless, within these constraints, it provides a useful basis for translating football tracking data into actionable performance insight.

Future work should focus on expanding the human-labeled dataset, improving event-tracking alignment, and exploring more advanced modeling approaches that can incorporate contextual tactical information, such as player positioning, defensive imbalance, and team structure. Nevertheless, within the scope of this project, the results provide a consistent and practically useful framework for analysing counter-attacks in elite football.

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